Dirac operators on Spin Manifolds

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Contents: We will define the group $\mathrm{Spin}(k)$ and spin manifolds. We will define the dirac operator on a spin manifold, called the Atiyah-Singer operator. Finally, we will give an application of the index theorem to a result regarding positive scalar curvature.

The group Spin(k)

The group Spin(k) is a compact Lie group. It is a double cover of SO(k), the group of rotations in $\mathbb{R}^n$.
Informally, the spin group lets us rotate space in a "complex" dimension.

Example

For $k = 1$, Spin(k) $=$ O(1) $= \{-1, 1\}$.
For $k = 2$, Spin(k) $=$ SO(2), the circle group. The double cover is just winding the circle around itself twice.
For $k = 3$, Spin(k) $= SU(2)$, the group of unit quaternions.

For the case $k = 2$, the spin group is not that interesting. However, for $k \geq 3$, the Spin(k) is simply connected, and is the univeral cover of SO(k).

Principal Bundles

Definition

Let X be a topological space and G be a topological group. A **principal G-bundle** is a fiber bundle $P \to X$, with a continuous action of G on P preserving the fibers. The fibers are *homeomorphic* to G, but not isomorphic.

A classical example of a principal bundle is the frame bundle of a smooth manifold M. The fiber of a point $m \in M$ is the set of frames for the tangent space at M. This is a principal GL(n)-bundle.

If M is oriented, we can similarly talk about the principal $\mathrm{SO}(n)$-bundle of positively oriented, orthogonal frames of an oriented manifold M.

Spin Manifolds

We can define a higher analogue of orientation using the spin group. Let M be an oriented Riemannian manifold of dimension k, and let P_{SO} be the principal $\mathrm{SO}(k)$-bundle of oriented orthonormal frames of the tangent bundle TM.

Definition

A **spin structure** on M is a principal $\mathrm{Spin}(k)$-bundle P_{Spin} over M, which is a double cover of P_{SO}, and such that the covering is compatible with the fiber-wise double cover $\mathrm{Spin}(k) \to \mathrm{SO}(k)$. if M admits a spin structure, it is called a **spin manifold**.

An example of a spin manifold is the torus $\mathbb{T}^n$. Its tangent bundle is trivial, so there are no obstructions to a spin structure.

Spinors

In the same way to every vector bundle there is an associated principal $\mathrm{GL}(k)$-bundle of frames, there is a way to associate to every principal G-bundle a vector bundle E, so that G encodes the data for how to piece together trivializations of E.

We will not go into details here, but to every spin manifold, we can associate a bundle S called the **spinor bundle**. This is an example of a **Clifford bundle**. Without going into details, these are bundles which have a **Clifford action**, meaning at every $x \in M$, tangent vectors to M at x act on the fiber S_x. This will be used to define Dirac operators.

There is a connection ∇ on the spinor bundle S called the **spin connection**. It can be thought of as a derivative on section of S, with

$$\nabla : C^\infty(S) \to C^\infty(T^*M \otimes S).$$

The Dirac Operator

let M be a spin manifold of even dimension n, S the spinor bundle, and ∇ the spin connection on S. The Dirac operator D on this clifford bundle is called the Atiyah-Singer operator.

Definition

We define D by the following sequence of maps

$$C^\infty(S) \to C^\infty(T^*M \otimes S) \to C^\infty(TM \otimes S) \to C^\infty(S),$$

where the first arrow is just the connection on S, the second arrow is the identification of T^*M with TM using the metric on M, and the last arrow comes from the clifford action of sections of TM on S fiberwise.

In a local orthonormal basis, we can write

$$Ds = \sum_i e_i \nabla_i s$$

where ∇_i is the connection applied to ∂_i.

Grading

The spinor bundle S is graded, meaning there is a decomposition

$$S = S^0 \oplus S^1$$

of S into even and odd parts. The clifford action on the spinor bundle is required to be chirality-reversing, meaning it sends elements of S^0 to S^1, and vice versa. From the local expression for the dirac operator D, we can see that D is an odd operator, meaning it reverses chirality. Thus, we can write

$$D = \begin{bmatrix} 0 & D^- \\ D^+ & 0 \end{bmatrix}.$$

Moreover, these operators are adjoints of one another. In particular, we have the expression

$$\mathrm{Index}(D^+) = \dim\ker D^+ - \dim\ker(D^+)^* = \dim\ker D^+ - \dim\ker D^-$$

for the index of D^+. Also, D is formally self adjoint.

The square of the Dirac operator Let e_i be an orthonormal frame at $m \in M$. Then we have

$$D^2 s = \sum_{i,j} e_j \nabla_j (e_i \nabla_i s) \tag{1}$$
$$= \sum_{i,j} e_j e_i \nabla_j \nabla_i s \tag{2}$$
$$= -\sum_i \nabla_i^2 s + \sum_{j<i} e_j e_i (\nabla_j \nabla_i - \nabla_i \nabla_j) s \tag{3}$$
$$= \nabla^* \nabla s + K s. \tag{4}$$

For a spin manifold M, this simplifies to

$$D^2 = \nabla^* \nabla + \frac{1}{4} \kappa,$$

where κ is the scalar curvature of M. This is known as the **Lichnerowicz formula**. The operator $\nabla^* \nabla$ is called the connection laplacian.

Twisting

We need an extra construction for our final theorem. If S is the spinor bundle on a spin manifold M, and E is any vector bundle on M, the vector bundle $S \otimes E$ is also a clifford bundle, meaning there is a clifford action on the fibers. We call this a **twisted spinor bundle**.

Here we define the **twisted Dirac operator** D_E. We have the more general formula

$$D_E^2 = \nabla^* \nabla + \frac{1}{4}\kappa + R^E,$$

where R^E only depends on the curvature of the bundle R^E and the dimension, and κ is the scalar curvature of M.

The comments about grading carry over to this case, so that we have the operators D_E^+ and D_E^-.

Enlargeable Manifolds

Definition

Let X be a Riemannian manifold. A map $f : X \to S^n$ is ε**-contracting** if we have

$$\|f^* v\| \leq \varepsilon \|v\|$$

for all tangent vectors v.
A compact riemannian manifold M of dimension n is **compactly enlargeable** if for every $\varepsilon > 0$, there exists an orientable Riemann covering space which admits an ε-contracting map to S^n of non-zero degree, such that the covering map is finite.

An example of an enlargeable spin manifold is the Storus $\mathbb{T}^n$ (*draw*).

Final Theorem

Theorem (Gromov-Lawson)

An enlargeable spin manifold may not carry a metric of positive scalar curvature.

Proof: Let M be an even dimensional compact spin manifold of even dimension. Suppose that M does carry a metric of positive scalar curvature, with $\kappa \geq \kappa_0 + c > 0$ for some κ_0 and $c > 0$.
Choose a complex vector bundle E_0 on the sphere S^n such that the top Chern class $c_n(E_0) \neq 0$. This is always possible.
Let $\varepsilon > 0$ and choose a finite orientable covering $\tilde{M} \to X$ which admits an ε-contracting map to the sphere of non-zero degree. Use f to pull back E_0 to a bundle E on $\tilde{X}$. Let S be the spinor bundle on $\tilde{M}$, and Consider the twisted Atiyah-Singer operator D_E on $S \otimes E$ as mentioned before. We have the formula

$$D_E^2 = \nabla^* \nabla + \frac{1}{4}\kappa + R^E.$$

We will now use (without proof) that the term R^E satisfies the estimate

$$\|R^E\| < \frac{1}{4}k_0$$

on each fiber of M. Let s be a section of $S \otimes E$. Then pointwise, we have

$$\langle D_E^2 s, s\rangle = |\nabla s|^2 + \frac{1}{4}\kappa|s|^2 + \langle R^E s, s\rangle. \tag{5}$$

For the last term, we have

$$\langle R^E s, s\rangle \leq \|R^E\||s|^2 < \frac{1}{4}k_0|s|^2,$$

so that

$$\langle D_E^2 s, s\rangle > |\nabla s|^2 + \frac{1}{4}\kappa|s|^2 - \frac{1}{4}|s|^2 > |\nabla s|^2 + c|s|^2.$$

This gives us the global L_2 estimate

$$\|D_E^2 s\| \geq c\|s\|.$$

Now it is clear that if $D_E s = 0$, we must have $s = 0$. In fact this implies $D_E^+ s = 0$, so that $\ker D^+ = \ker D^- = \{0\}$. In particular, we have

$$\mathrm{Index}(D_E^+) = \dim\ker D^+ - \dim\ker D^- = 0.$$

However, from a calculation using the Index formula for twisted dirac operators, we can show the index of the dirac operator is non-zero. This gives a contradiction, and thus X may not admit a metric of positive scalar curvature.